Biomedical Signal Processing and Machine Learning

Prof. Stefano Diciotti's Module – Practical part

Second Cycle Degree/Two-Year Master's in Biomedical Engineering

8 June 2026

Setting up the development platform (Google Colab)

1. Go to eol.unibo.it, log in to today's session, and download the file **'Template_BSPML_20260608.ipynb'.**
2. Go to https://colab.research.google.com/ , and log in at the top right with your Google account.
3. Upload the template to Google Colab, for example, by selecting **'File -> Upload Notebook -> Upload'**

Documentation

Directly use the following links for the Python documentation (DO NOT use Google or other search engines because of firewall settings):

- https://pandas.pydata.org/docs/
- https://scikit-learn.org/stable/
- https://numpy.org/doc/
- https://matplotlib.org/stable/index.html

Exercise: Hyperparameter optimization with nested cross-validation

You are provided with a synthetic biomedical dataset generated using make_classification(). The dataset contains 500 samples and 20 features, representing a binary classification problem (e.g., healthy vs. diseased patients).

Your task is to build and evaluate a machine learning model using Logistic Regression and nested cross-validation.

Requirements:

1. Use a Logistic Regression classifier.
2. Define a hyperparameter grid for the regularization parameter C with the following values:
    - 0.01
    - 0.1
    - 1
    - 10

    - 100

3. Implement an inner 5-fold cross-validation loop to optimize hyperparameters using GridSearchCV.
4. Implement an outer cross-validation loop using cross_validate with 5-fold cross-validation to estimate the model's generalization performance.
5. Use ROC AUC as the optimization metric during hyperparameter tuning.
6. Evaluate the final model using the following metrics:
    - accuracy
    - ROC AUC
7. Report:
    - The mean and standard deviation of accuracy across the outer folds.
    - The mean and standard deviation of ROC AUC across the outer folds.

Questions (please answer in a text cell):

1. Why is nested cross-validation preferred over a single cross-validation procedure when hyperparameters must be optimized?
2. Which metric (accuracy or ROC AUC) is more informative for this biomedical classification problem? Explain your answer.

Additional analysis: effect of class imbalance

The initial dataset has a class distribution defined by:

weights=[0.6, 0.4]

Repeat the analysis with increasing levels of class imbalance:

weights=[0.8, 0.2]

weights=[0.96, 0.04]

For each dataset:

1. Perform the same nested cross-validation procedure described above.
2. Report the mean Accuracy and mean ROC AUC obtained across the outer folds.
3. Compare the results obtained for the three datasets.

Discussion Questions (please answer in a text cell)

1. How does increasing class imbalance affect accuracy and ROC AUC?
2. Which metric appears to be more robust to class imbalance?
3. In a biomedical diagnosis setting, would accuracy alone be sufficient to evaluate model performance? Why or why not?

Submitting the solution on eol.unibo.it

Save your Python code on your PC, rename it, and upload it to eol.unibo.it